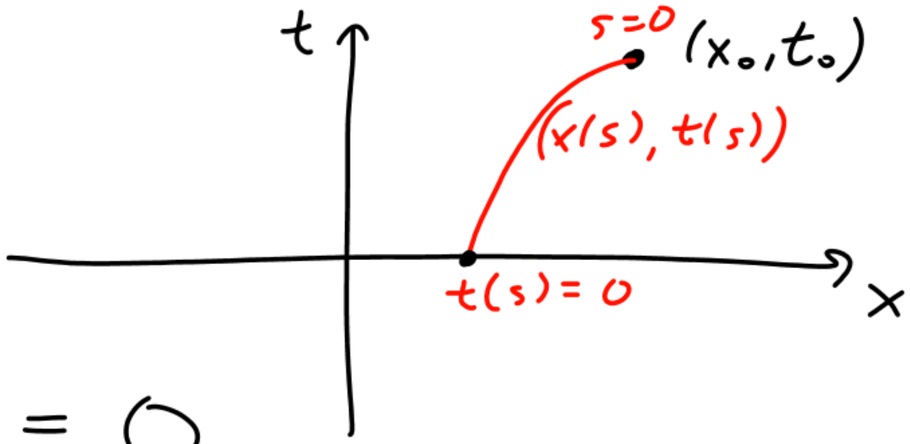
Test 2 : Wed, Nov 5 in class

Covers: Heat Equation, Wave Equation,
Fourier Transform

Practice Test is available

HW 4 due Wed, Oct 29

# General transport equations




Variable-coefficient transport:

$$\boxed{a(x,t)}u_t + \boxed{b(x,t)}u_x + c(x,t)u = 0$$

How to find characteristic ODEs?

$$z(s) = u(x(s),t(s))$$

$$\frac{d}{ds}z(s) = u_x\boxed{\frac{dx}{ds}} + u_t\boxed{\frac{dt}{ds}} = au_t + bu_x = -c(x,t)u$$

Chain Rule

$$\boxed{\begin{aligned}\frac{dx}{ds} &= b(x(s),t(s))\\ \frac{dt}{ds} &= a(x(s),t(s))\end{aligned}}$$

$$\rightarrow \boxed{\frac{dz}{ds} = -c(x(s),t(s))z(s)}$$

$$\boxed{1}\begin{cases} u_t + \boxed{3t^2}\, u_x = 2tu \\ u(x,0) = g(x) \end{cases}$$

$$z(s) = u(x(s), t(s))$$

$$\frac{d}{ds} z(s) = u_x \boxed{\frac{dx}{ds}} + u_t \boxed{\frac{dt}{ds}}$$

Char. ODEs
$$\begin{cases} \dfrac{dt}{ds} = 1 \\ \dfrac{dx}{ds} = 3t^2 \end{cases}$$

$$\frac{dt}{ds} = 1, \quad t = s + c$$

$$t(0) = t_0 = c$$

$$\longrightarrow \underline{t = s + t_0}$$

$$s = t - t_0$$

$$\frac{dx}{ds} = 3t^2 = 3(s+t_0)^2$$

$$x(s) = (s+t_0)^3 + c$$

$$s = 0, \quad x(0) = x_0 \qquad x_0 = t_0^3 + c$$

$$c = x_0 - t_0^3$$

$$\longrightarrow \underline{x(s) = (s+t_0)^3 + x_0 - t_0^3}$$

PDE: $\dfrac{dz}{ds} = 2tz = 2(s+t_0)\, z(s)$

$$\frac{dz}{ds} = 2(s+t_0)z$$

$$\frac{1}{z}\frac{dz}{ds} = 2(s+t_0)$$

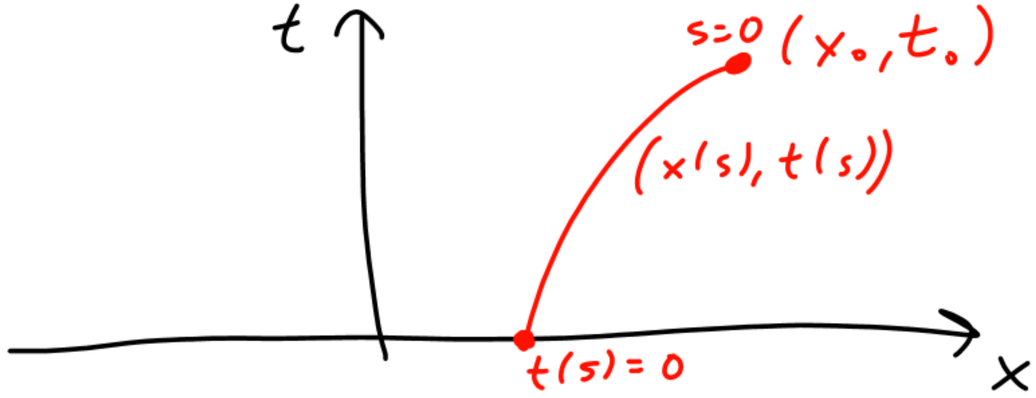
$$\ln(z) = (s+t_0)^2 + c$$

$$z(s) = e^{(s+t_0)^2} e^{c}$$

$s=0$: $\quad u(x_0,t_0) = e^{t_0^2} e^{c}$

$$e^{c} = e^{-t_0^2} u(x_0,t_0)$$

$$z(s) = u(x_0,t_0)\, e^{-t_0^2} e^{(s+t_0)^2}$$




$$z(s) = u(x(s), t(s))$$

$$= u\left((s+t_0)^3 + x_0 - t_0^3,\ t_0 + s\right)$$

$$z(-t_0) = u(x_0 - t_0^3, 0) = g(x_0 - t_0^3)$$

$$z(-t_0) = u(x_0,t_0)\, e^{-t_0^2}$$

$$u(x_0,t_0) = g(x_0 - t_0^3)\, e^{t_0^2}$$

$$\boxed{u(x,t) = g(x - t^3)\, e^{t^2}}$$

$$\begin{cases} \boxed{x}\, u_t \boxed{-t}\, u_x = u, & x > 0,\ t > 0 \\ u(x,0) = g(x) \end{cases}$$

$$z(s) = u(x(s), t(s))$$

$$\frac{dz}{ds} = u_x \boxed{\frac{dx}{ds}} + u_t \boxed{\frac{dt}{ds}}$$

Char. ODEs:

$$\begin{cases} \dfrac{dx}{ds} = -t \\ \dfrac{dt}{ds} = x \end{cases}$$

Write this ODE system in matrix form:

$$\frac{d}{ds}\begin{bmatrix} x \\ t \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x \\ t \end{bmatrix}$$

Find eigenvalues of $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$:

$$\begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0 \qquad \lambda = \pm i$$

$$\left(\text{Solutions } \begin{bmatrix} x \\ t \end{bmatrix} \sim e^{Re(\lambda)s}\, e^{Im(\lambda)s} \sim e^{\pm is} \sim \cos(s) \text{ or } \sin(s)\right)$$

Guess

$$x(s) = c_1 \cos(s) + c_2 \sin(s)$$
$$t(s) = c_3 \cos(s) + c_4 \sin(s)$$

$$\frac{dx}{ds} = -c_1 \sin(s) + c_2 \cos(s)$$
$$= -c_3 \cos(s) - c_4 \sin(s)$$

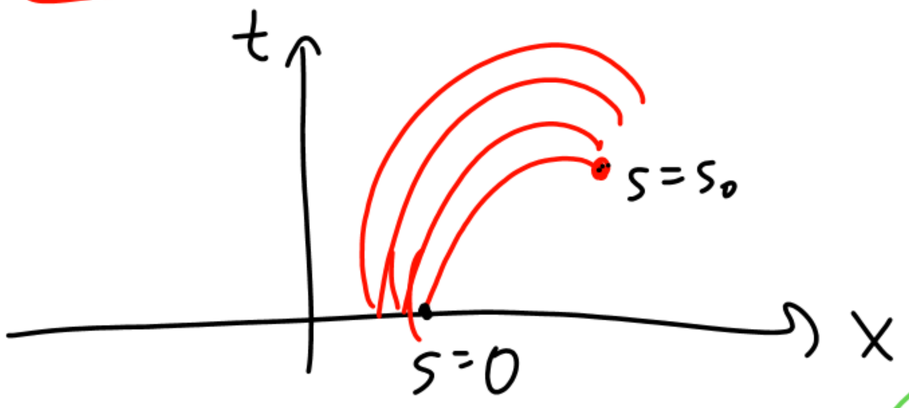
$$\rightarrow \quad -c_3 = c_2 \,, \quad c_4 = c_1$$

$$x(s) = c_1 \cos(s) + c_2 \sin(s)$$
$$t(s) = -c_2 \cos(s) + c_1 \sin(s)$$

Check: $\frac{dt}{ds} = c_2 \sin(s) + c_1 \cos(s)$
$= x$ ✓

We will choose $c_2 = 0$

$$\rightarrow \boxed{\begin{aligned} x(s) &= c \cos(s) \\ t(s) &= c \sin(s) \end{aligned}}$$ Char. curves



PDE:

$$x u_t - t u_x = u$$

$$\frac{dz(s)}{ds} = x(s) u_t - t(s) u_x = z(s)$$

$$\rightarrow z(s) = z(0) e^s$$

$$z(0) = \cancel{u(x_0, t_0)} \quad u(x(0), 0)$$

$$\rightarrow z(s) = u(c, 0) e^s$$

Which s-value corresponds to $t(s) = 0$? $\rightarrow s = 0$

$$x(s) = c\cos(s)$$
$$t(s) = c\sin(s)$$

$s = 0$: $\quad x(0) = c$
$\quad t(0) = 0$

Which s-value gives $(x_0, t_0)$?

$$x_0 = c\cos(s)$$
$$t_0 = c\sin(s)$$

$$x_0^2 + t_0^2 = c^2(\cos^2(s) + \sin^2(s)) = c^2$$

$$\boxed{c = \sqrt{x_0^2 + t_0^2}}$$

$$\frac{t_0}{x_0} = \frac{c\sin(s)}{c\cos(s)} = \tan(s)$$

$$\boxed{s = \arctan\left(\frac{t_0}{x_0}\right)}$$

Char. curves:

$$x(s) = \sqrt{x_0^2+t_0^2}\cos(s)$$
$$t(s) = \sqrt{x_0^2+t_0^2}\sin(s)$$

$$(x(0), t(0)) = (\sqrt{x_0^2+t_0^2}, 0)$$

Let $s_0 = \arctan\left(\frac{t_0}{x_0}\right)$

$$(x(s_0), t(s_0)) = (x_0, t_0)$$

$$z(s) = z(0)e^s$$
$$= u(x(0), t(0))e^s$$
$$= u(\sqrt{x_0^2+t_0^2}, 0)e^s$$

$$z(s) = g(\sqrt{x_0^2+t_0^2})e^s$$

$$z(s_0) = g(\sqrt{x_0^2+t_0^2})e^{s_0}$$

$$u(x_0, t_0) = g(\sqrt{x_0^2+t_0^2})e^{\arctan\left(\frac{t_0}{x_0}\right)}$$

$$u(x,t) = g(\sqrt{x^2+t^2})e^{\arctan\left(\frac{t}{x}\right)}$$

Nonlinear example:

$$\begin{cases} u_t + u_x = u^2 \\ u(x,0) = g(x) \end{cases}$$

Characteristic ODEs:

$$z(s) = u(x(s), t(s))$$

$$\frac{dz}{ds} = u_x \frac{dx}{ds} + u_t \frac{dt}{ds} = u^2$$

$$\frac{dx}{ds} = 1 \longrightarrow \boxed{x(s) = x_0 + s}$$

$$\frac{dt}{ds} = 1 \longrightarrow \boxed{t(s) = t_0 + s}$$

$$\frac{dz}{ds} = z^2$$

$$\frac{1}{z^2}\frac{dz}{ds} = 1$$

$$-\frac{1}{z} = s + c$$

$$s = 0: \quad \frac{-1}{z(0)} = c \longrightarrow c = \frac{-1}{z(0)}$$

$$-\frac{1}{z} = s - \frac{1}{z(0)} \longrightarrow z(s) = \frac{-1}{s - \frac{1}{z(0)}}$$

$$\boxed{z(s) = \frac{1}{\frac{1}{z(0)} - s}}$$

$$z(0) = u(x_0, t_0)$$

$$z(s) = \frac{1}{\frac{1}{u(x_0,t_0)} - s}$$

$$z(s) = \frac{1}{\frac{1}{u(x_0,t_0)} - s}$$

$$z(-t_0) = \frac{1}{\frac{1}{u(x_0,t_0)} + t_0}$$

$$= u(x_0 - t_0, 0) = g(x_0 - t_0)$$

$$g(x_0 - t_0) = \frac{1}{\frac{1}{u(x_0,t_0)} + t_0}$$

$$\frac{1}{u(x_0,t_0)} + t_0 = \frac{1}{g(x_0 - t_0)}$$

$$\boxed{u(x_0,t_0) = \frac{1}{\frac{1}{g(x_0-t_0)} - t_0}}$$

This solution may blow up in finite time.

If $g(x) \leq M$ for all $x$,

$$\frac{1}{g(x-t)} - t \geq \frac{1}{M} - t > 0$$

if $t < \frac{1}{M}$

$\rightarrow$ Solution exist up to time $\frac{1}{M}$.

What is a nonlinear PDE?

(Example: $L = -\Delta$, $D = C^2(\mathbb{R}^n)$)

<u>Def</u> A differential operator $L$, with domain $D$, is <u>linear</u> if

i) $L(af) = aLf$ for all $a \in \mathbb{R}$, $f \in D$.

ii) $L(f+g) = Lf + Lg$ for all $f, g \in D$.

e.g. $L = \frac{\partial}{\partial t} + a(x)\frac{\partial}{\partial x}$

$$L(f+g) = \left(\frac{\partial}{\partial t} + a(x)\frac{\partial}{\partial x}\right)(f+g)$$

$$= \frac{\partial}{\partial t}(f+g) + a(x)\frac{\partial}{\partial x}(f+g) = f_t + g_t + af_x + ag_x$$

$$= Lf + Lg \quad \checkmark$$

A <u>Linear PDE</u> is a PDE of the form

$$\underline{Lu = f}$$

for a linear operator $L$ and a function $f$ (possibly $0$) that is independent of $u$.

Any PDE that is not linear is <u>nonlinear</u>.